

02YMECHZ - Mechanics

Exam

6.2.2026 / 10:00 - 12:00

Q1: Calculate the gravitational potential and gravitational field due to a uniform ring of radius a and mass m at a point on the axis of the ring a distance z from its center. Are your results consistent with the limits for $z \gg a$ and $z = 0$? Justify your answer. — 1 point

Q2: A rigid body consists of two identical point masses, each of mass m , connected by a massless rod. They are located at fixed positions $\mathbf{r}_1 = (a, a, 0)$ and $\mathbf{r}_2 = (-a, -a, 0)$ in a Cartesian coordinate system. The body rotates with a constant angular velocity $\boldsymbol{\omega} = (\omega, 0, 0)$ about the x -axis.

- a. Compute the inertia tensor $\mathbf{I}$ of the system in the (x, y, z) basis. — 0.4 point
- b. Compute the total angular momentum vector $\mathbf{L}$ of the body. — 0.4 point
- c. Find the angle θ between $\boldsymbol{\omega}$ and $\mathbf{L}$. — 0.2 point

Given: The components of the inertia tensor: $I_{ij} = \sum_k m_k (\delta_{ij} r_k^2 - r_{k,i} r_{k,j})$.

Q3: The equation of motion for a driven oscillator is given as $m\ddot{x} + b\dot{x} + kx = F_0 \cos(\omega t)$, where m is the mass, b is the damping coefficient, k is the spring constant, and $F_0 \cos(\omega t)$ is the driving force. The solution for the displacement $x(t)$ of the oscillator is given by $x(t) = x_c(t) + x_p(t)$, where $x_c(t)$ is the complementary solution and $x_p(t) = A \cos(\omega t - \delta)$ is the particular solution.

- a. Describe the oscillator's behavior in the long-time limit, i.e., $t \rightarrow \infty$. — 0.3 point
- b. What is the amplitude of oscillation in terms of k and F_0 when the system is driven with frequency $\omega \ll \omega_0$? What is the physical meaning of this? — 0.3 point
- c. What is the amplitude of oscillation in terms of k , F_0 and Q when the system is driven in resonance frequency? Compared to what you found in part **b.** and considering the fact that for most of the oscillators $Q \gg 1$, what does driving the system with the resonance frequency do to the amplitude? — 0.4 points

Given: $A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\frac{b\omega}{m})^2}}$, $Q = \frac{m\omega_0}{b}$, $\omega_0 = \sqrt{\frac{k}{m}}$

Q4: A child stands on the center of a horizontal merry-go-round that rotates with constant angular velocity $\boldsymbol{\Omega} = \Omega \hat{z}$. The surface is frictionless. At time $t = 0$, a child gently pushes a small puck of mass m so that, relative to the rotating platform, the puck's initial velocity is purely radial and directed outward. Work entirely in the rotating (non-inertial) frame of the merry-go-round.

- a. Write the equation of motion of the puck in the rotation frame including all fictitious (inertial) forces. — 0.4 point
- b. Identify explicitly the centrifugal and Coriolis forces acting on the puck and explain qualitatively why the puck's trajectory does not remain purely radial. — 0.3 point

- c. Determine the direction of the deflection (clockwise or counterclockwise when viewed from above) and describe the trajectory of the puck qualitatively as it moves outward.
— *0.3 point*

Given: Coriolis force $\vec{F}_c = -2m\vec{\Omega} \times \vec{v}$, Centrifugal force $\vec{F}_{cf} = -m\vec{\Omega} \times (\vec{\Omega} \times \vec{r})$.